MILITARY SPECIFICATION

GAGE SYSTEM, FUEL-QUANTITY, CAPACITOR-TYPE, NONVACUUM TUBE, INTEGRALLY LIGHTED, GENERAL SPECIFICATION FOR

This specification has been approved by the Department of the Air Force and by the Navy Bureau of Aeronautics.

1. SCOPE

1.1 Scope.- This specification covers the general requirements for capacitor-type fuel-quantity gages which do not utilize electron tubes.

1.2 Classification.- Gages shall be of the following types, as specified (see 6.2):

(a) Type I - (uncompensated)
(b) Type II - (compensated)

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)
QQ-Z-325	Zinc Plating (Electrodeposited)

Military

MIL-P-116	Preservation, Methods of
MIL-S-3136	Standard Test Fluids Hydrocarbon and Iso-Octane
MIL-T-4687	Tester, Capacitor Type Fuel Quantity Gage Tank Unit, Type MD-2A
MIL-C-5015	Connectors, Electric, "AN" Type
MIL-D-5028	Drawings and Data Lists: Preparation of Manufacturers' (for Production Aeronautical and Associated Equipment)
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-G-5572	Gasoline, Aviation: Grades 80/87, 91/96, 100/130, 115/145
MIL-F-5624	Fuel, Aircraft Turbine and Jet Engine Grades JP-3, JP-4, and JP-5

FSC 6680

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MIL-P-5633	Packaging and Packing of Aircraft Material in Steel Shipping Containers
MIL-P-6064	Packaging of Lightweight Aircraft Accessories
MIL-I-6181	Interference Limits, Tests and Design Requirements, Aircraft Electrical and Electronic Equipment
MIL-L-6880	Lubrication of Aircraft, General Specification for
MIL-P-6906	Plates, Information and Identification
MIL-W-7139	Wire, Electrical, Insulated, Copper, 600-Volt, 400° Fahrenheit
MIL-S-7742	Screw Threads, Standard, Aeronautical
MIL-E-7894	Electric Power, Aircraft, Characteristics of
MIL-G-7940	Gages, Fuel Quantity, Capacitor Type, Installation and Calibration of
MIL-T-8579	Tester, Fuel Quantity Gage, Variable Capacitance, Field, Type MD-1
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-C-8721	Cable, Radio Frequency, Coaxial, Miniature
MIL-W-16878	Wire, Electrical (Insulated, High Temperature)
MIL-L-25467	Lighting, Integral, Instrument, General Requirements for
MIL-C-25516	Connectors, Electrical, Miniature, Shielded or Unshielded, Environmental Resistant Type, General Specification for

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MS33516	Case - Standard Dimensions for 3-1/4 Inch Size Instrument
MS33554	Case, Instrument, 1-1/2 Dial, Standard Dimensions for

DRAWINGS

Air Force-Navy Aeronautical Standard Drawings

AN4130	Fitting - Synthetic Tank Single Plane Flush and Recessed
AND10398	Metals, Definition of Dissimilar
AND10412	Case - 2-Inch Size Instrument, Standard Dimensions for

PUBLICATIONS

Air Force-Navy Aeronautical Bulletins

No. 143	Specifications and Standards; Use of
No. 405	Storage Life - Aeronautical Articles

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Qualification.- The gage systems furnished under this specification shall be a product which has been tested and has passed the Qualification tests specified herein.

3.2 Component parts.- The 2-unit fuel quantity gage system shall consist basically of fuel tank probes, intermediate devices and compensators if necessary, individual indicators, and totalizing indicators. The 3-unit system shall consist basically of fuel tank probes, intermediate devices, individual indicators, and totalizing indicators.

3.3 Materials.-

3.3.1 Protective treatment.- When materials are used in the construction of the gage that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.3.2 Materials shall conform to applicable specifications as specified herein. When materials are used which are not specifically designated, they shall be entirely suitable for the purpose. The use of lightweight materials and weight-saving designs is a major consideration, and their use shall be investigated and exploited to the greatest possible extent.

3.3.3 Metals.- Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion in fuels, salt spray, atmospheric conditions, aerodymanic heating and loading conditions to which the gage will be subjected when in storage or during normal service use.

3.3.4 Dissimilar metals.- Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined by Drawing AND10398.

3.3.5 Nonmagnetic materials.- Nonmagnetic materials shall be used for all parts, except where magnetic materials are essential.

3.3.6 Fungus-proof materials.- Materials which are not nutrients for fungi shall be used to the greatest extent practicable. Where materials that are nutrients for fungi must be used, such materials shall be either treated with a fungicidal agent or protected by a hermetically sealed enclosure.

3.3.7 Corrosive fumes.- Materials used in the interior construction of the system components shall not produce corrosive or deleterious fumes under the test conditions specified herein.

3.3.8 Neither zinc nor magnesium metal or plating shall be used in contact with fuel.

3.3.9 Selection of materials.- Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with ANA Bulletin No. 143, except as provided in the following paragraph.

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3.3.9.1 Standard parts.- Standard parts (MS, AN, or JAN) shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc, may be used, provided they possess suitable properties and are replaceable by the standard parts (MS, AN, or JAN) without alteration, and provided the corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings. In the event there is no suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.4 Design and construction.- The basic gage shall be designed to indicate the quantity of usable fuel in a given tank by measurement of the electrical capacitance between two or more electrodes of an electrostatic capacitor (tank unit) mounted in the fuel tank. The design shall provide linear indication. The gage shall consist of one or more tank units and an indicator. The indicator shall consist of an indicating unit and an amplifier packaged as a single unit or as separate units. The system design shall be a null-balancing, 3-wire capacitance bridge, and shall provide linear indication. The design shall also provide an output signal for totalizing purposes, if required for specific applications as directed by the cognizant military activity. Incorporation of the added totalizing feature shall entail the inclusion of a suitable summing device and an indicating unit which shall be designed to indicate the total amount of fuel registered by the individual gages. This summing device may be incorporated within the indicating unit. The tank unit, or units, shall be characterized such that a constant capacitance is registered for each unit volume of fuel sensed.

3.4.1 Uncompensated gage.- The basic uncompensated gage shall consist of one or more tank units and an indicator.

3.4.2 Compensated gage.- The basic compensated gage shall consist of one or more tank units, a compensator, and an indicator. The compensator sensing unit may be an individual mounted component. Compensation shall be based on the value of the dielectric constant of the fuel and shall indicate the mass of fuel when the dielectric constant and density are related in accordance with the following formulas:

(a) If jet fuel, type JP-4, is normally the only fuel used in the fuel tank, the following formula shall be used:

$$\frac{K - 1}{D} = 0.12192 \left[1 + 0.3373 (K-1)\right] \qquad (1)$$

(b) If jet fuel, type JP-4, and octane fuels, types 91/96, 100/115, or 115/145, are normally to be used interchangeably, the following formula shall be used:

$$\frac{K - 1}{D} = 0.12700 \left[1 + 0.28953 (K-1)\right] \qquad (2)$$

where K is the dielectric constant and D is the density in lb/gal.

3.4.3 Fuel characteristics.- Nominal fuel and end point fuel as specified herein shall be fuels having the characteristics listed in table I.

TABLE I

Fuel characteristics

Characteristics	Uncompensated gage	JP-4 only	JP-4 and octanes
Nominal dielectric constant	1.990	2.090	2.041
Nominal density	6.000	6.539	6.299
End point dielectric constant	2.088	2.245	2.242
End point density	6.593	7.191	7.191

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3.4.4 The gages shall be so constructed that no parts will work loose in service. They shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service. Insofar as practicable, pivots, bearings, and gears shall neither bind nor shake and shall be frictionless.

3.4.5 The components shall be so constructed that adjustments and repairs can be made easily by personnel of overhaul depots.

3.4.6 Fire hazard.- The electrical circuit of the gage system shall be so designed that the value and magnitude of the potentials applied to the tank unit, or units, shall not create a fire hazard in the airplane fuel tank.

3.4.6.1 The design of the circuit shall be such that the current in the leads running to the sensing electrodes of tank units and compensator sensing units shall in no case exceed 0.2 ampere. This requirement shall apply both to normal operating conditions and to any combination of the following abnormal conditions:

(a) Short circuiting of any combination of electrodes in the tank unit or compensator sensing unit.

(b) Short circuiting of any capacitor or combination of capacitors elsewhere in the bridge circuit.

(c) All switches and other adjustments in the bridge circuit set to any desired position.

3.4.7 General.- The type II compensated gage system shall be applicable to all types of aircraft installations wherein the fuel used is type JP-4 of Specification MIL-F-5624, or wherein JP-4 and octane fuels are used interchangeably. The type I noncompensated gage system shall be applicable to installations utilizing aviation fuels 80, 91/96, 100/130, and 115/145, of Specification MIL-G-5572.

3.4.8 Component parts.-

3.4.8.1 Totalizer.- The totalizer shall consist basically of an indicator and a summing device. The summing device may be incorporated within the totalizing indicator. The individual gages associated with the totalizing gage system shall be designed to provide suitable electrical output signals for summation by the totalizer. Malfunction of the totalizer shall not adversely affect the performance of the individual gages.

3.4.8.2 Devices.- Various devices may be used to perform auxiliary functions, such as simulating empty capacitance of tank units, center of gravity control, etc. These extra components which are an integral part of the fuel-quantity gage system or any of its components shall meet the general requirements of this specification.

3.4.8.3 The number of component parts of the fuel gage system shall be kept to the minimum number consistent with good engineering practice.

3.4.8.4 Tank units.- The basic tank unit shall consist of a suitable capacitative arrangement of conductors, such as a pair of concentric cylinders electrically insulated from each other and forming a 3-wire capacitor. The design shall be such that the tank unit can be disassembled and reassembled without the use of special tools, except as approved by the cognizant military activity. It shall be so constructed as to permit fuel into which the unit has been immersed to rise freely to its natural level between the conductors, to insure that up to this level, the fuel will serve as the dielectric substance. Above the level of the fuel, a mixture of air and vapor will be the dielectric substance. Nonsensing portions of the sensing element shall not exceed the following

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dimensions: 0.600 inch at the flange end and 0.400 inch at the opposite end for externally top-mounted and flush-mounted tank units; and 0.400 inch at both ends for internally mounted tank units. Nonsensing portions of bottom externally mounted units at the flange end shall be subject to the approval of the cognizant military activity. It shall be so constructed as to preclude the formation of deposits and sediments between the conductors. The minimum allowable distance between the conductors shall be 0.125 inch. The tank unit shall be designed to make electrical connections readily accessible for testing the tank unit, using the MD-2 field tester of Specification MIL-T-4687 and,to make calibration adjustments, using the MD-1 field tester of Specification MIL-T-8579. The materials used in the construction of the tank units shall be impervious to aircraft fuels.

3.4.8.4.1 Characterized tank units.- Characterized tank units shall be designed to provide a linear relationship between the electrical capacitance and the volume of fuel sensed. The basic design shall be suitable for producing high capacitance ratios (i.e., large ratio between the maximum and minimum capacitance per unit length) and following calibration curves with a high degree of accuracy. The basic design shall possess good production adaptability. The size and weight shall be kept to the minimum consistent with high-quality aircraft instrument design.

3.4.8.4.2 Mounting types.- The tank unit designs shall include externally mounted types, internally mounted types, and flush-mounted types, as required for specific installations. The units shall be suitable for installation in accordance with Specification MIL-G-7940.

3.4.8.4.3 Mounting flange.- The external top-mounted, and bottom-mounted types, when specifically authorized by the cognizant military activity, shall be designed and constructed for installation in a fuel tank fitting which has dimensions conforming to Drawing AN4130. Flange-mounted tank units shall have flanges conforming to size 4, 5, or 6. The means provided for mounting the other types shall be entirely suitable for the purpose and shall be subject to approval by the cognizant military activity. Flush-mounted types of tank units shall be entirely suitable for the conditions of flexure, stress, and temperature encountered in the aircraft.

3.4.8.4.4 Connector housing.- For externally mounted type tank units, the projection of the connector housing beyond the flange surface shall be held to a minimum practicable distance in order to facilitate mounting.

3.4.8.4.5 Seal.- The externally and flush-mounted-type-tank unit head shall be sealed to prevent the leakage of fuel from the inside of the tank to the outside.

3.4.8.4.6 Compensator sensing unit.- The compensator sensing unit shall meet the same general requirements as the tank unit. It shall be designed to extend a minimum distance from the bottom of the fuel tank in which it will be installed. The unit shall be designed to make electrical connection readily accessible for testing the unit, using the MD-2 field tester of Specification MIL-T-4687 and,to make calibration adjustments, using the MD-1 field tester of Specification MIL-T-8579. The compensator sensing unit may be an integral part of the tank unit or may be a separate component. When the compensator is a separate component or an integral part of the bottom externally mounted tank unit, the distance from the flange to the lowest sensing level of the compensator shall be suitable for the specific application and subject to the approval of the cognizant military activity. Paragraphs 3.4.8.4.3, 3.4.8.4.4, and 3.4.8.4.5 shall apply to flange-mounted compensator sensing units which are separate components.

3.4.8.4.7 Standard electronic parts.- Electronic parts and the application thereof shall be in accordance with Specification MIL-E-5400.

3.4.8.4.7.1 Electronic nonstandard parts.- All nonstandard parts used for amplification, rectification, and temperature compensation shall be entirely suitable for the purpose and shall be subject to the approval of the activity responsible for qualification.

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3.4.8.5 Indicator.-

3.4.8.5.1 General.- The indicator shall be an electrical device which continuously indicates the quantity of fuel in pounds. It shall consist of an indicating unit and an intermediate device packaged as a single unit or as separate units. If packaged separately, both units shall be hermetically sealed. Provisions shall be made in the design to permit the addition of level warning switching mechanisms and output signal controls, such as potentiometers as required for specific applications. The indicator designs shall conform to the dial presentation types shown on figures 1 through 5. All indicators shall be designed to provide readily accessible connections for testing the operation of the gage system and making calibration adjustments, using the MD-1 field tester of Specification MIL-T-8579. If stops are provided, they shall not interfere with the calibration adjustments or with the normal operation of the gage.

3.4.8.5.2 Speed of response.- The indicator shall register from empty to full or vice versa in less than 90 seconds and more than 15 seconds. The response time for all indicators shall be of sufficient length to prevent oscillation in turbulent air.

3.4.8.5.3 Indicator case.- The case design shall be in accordance with those shown on figure 1, 2, and 4. The case shall be constructed of a suitable material applicable to hermetic sealing, and the material shall be uniform in texture with a smooth external surface. The case shall provide a hermetically sealed enclosure for the internal electrical devices and mechanism. The design of the case shall be such that the internal devices and mechanism may be removed from the case, replaced, and the case resealed at least three times. This shall be accomplished without the use of special tools and fixtures, unless they are approved by the cognizant military activity. The hermetic sealing shall be so accomplished that the seal will not be dependent upon material which will be affected by the action of any atmosphere to which the case may be subjected. The indicator case shall be the minimum length consistent with high-quality instrument design practice and shall not exceed the maximum lengths specified in table II. The length limitation may be modified only with the specific written approval of the activity responsible for qualification.

TABLE II

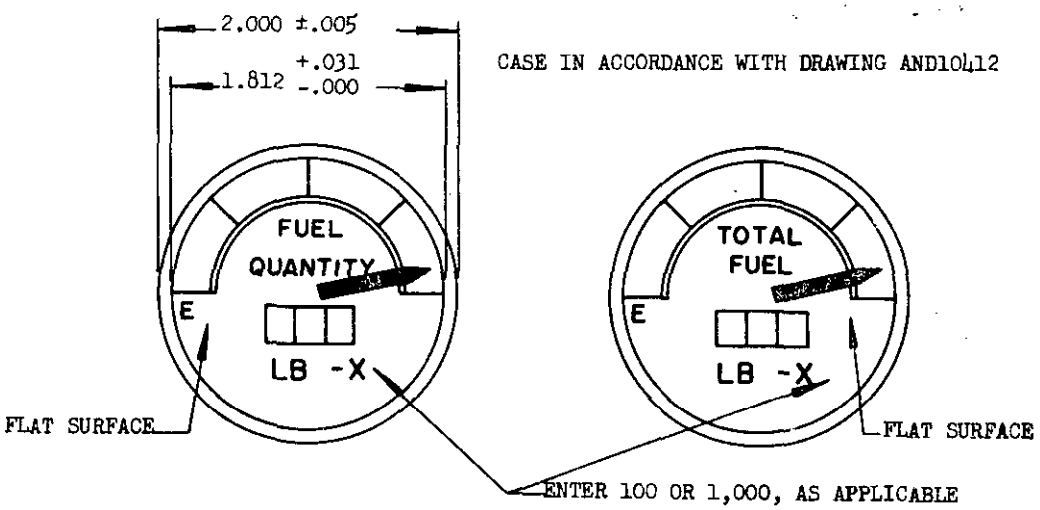
Case lengths

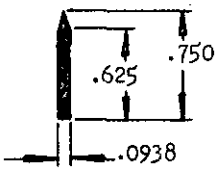
Diameter (inches)	Type of system	Maximum length (inches)
1-1/2	3 unit	4
2	2 unit	6
2	3 unit	5
2 (2 servos)	3 unit	6
2 (2 servos)	2 unit	6-1/2

3.4.8.5.3.1 The filling medium shall be of at least 98-percent purity, free of dust particles, and containing not more than 0.006 milligram of water vapor per liter (dewpoint -65°C) at the filling pressure. The filling medium shall be either helium or a mixture of from 88 to 92 percent nitrogen at the remainder helium. When practicable, the 100 percent helium shall be used. The absolute pressure of the filling medium in the case shall be approximately 1 atmosphere.

3.4.8.5.3.2 Indicator lighting.- The indicator shall be integrally lighted. The lighting system shall be in accordance with Specification MIL-L-25467.

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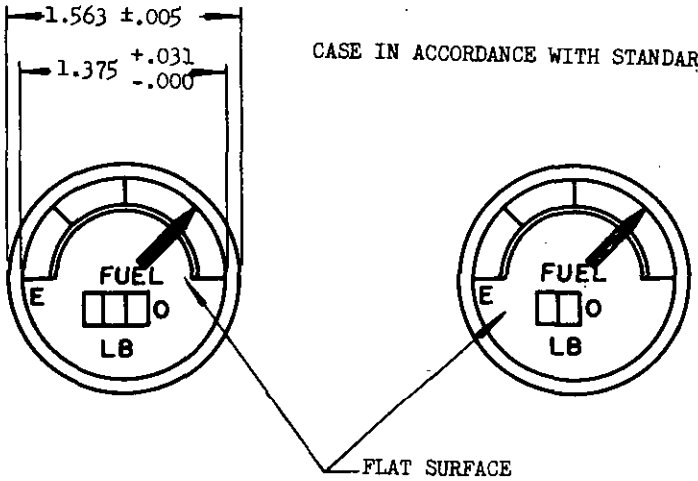
POINTER

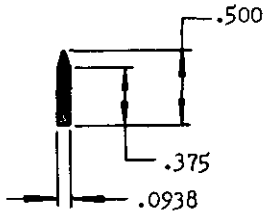
MARKINGS	HEIGHT OR LENGTH ±.010	WIDTH OF LINE OF GRADUATION ±.005
FUEL QUANTITY	.156	.031
TOTAL FUEL	.156	.031
LB -X	.140	.025
E	.200	.031
MAJOR GRAD	.250	.031
COUNTER NUMERALS	.200	.031

DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: ANGLES ±0°15', DECIMALS ±0.005.

FIGURE 1. Dial - 2-inch size, 180-degree scale

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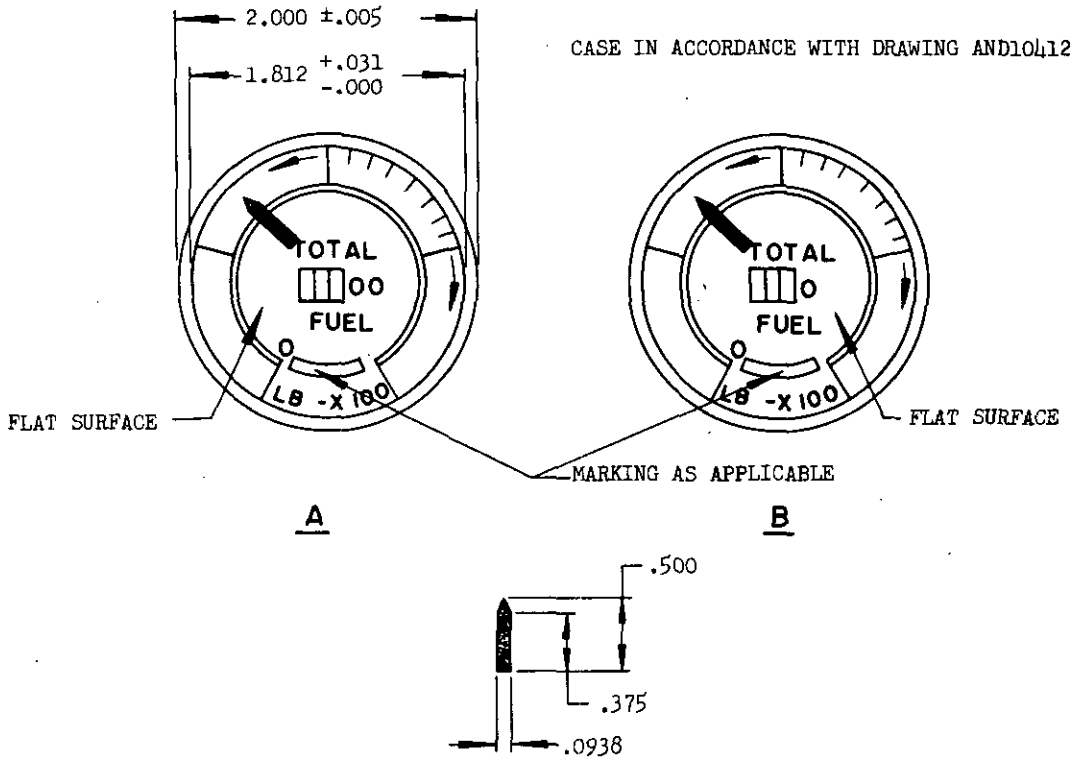
POINTER

MARKINGS	HEIGHT OR LENGTH ±.010	WIDTH OF LINE OR GRADUATION
FUEL	.156	.031
LB	.140	.025
E	.200	.031
MAJOR GRAD	.188	.031
COUNTER NUMERALS	.180	.031

DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: ANGLES ±0°15', DECIMALS ±0.005.

FIGURE 2. Dial - 1-1/2 inch size, 180-degree scale

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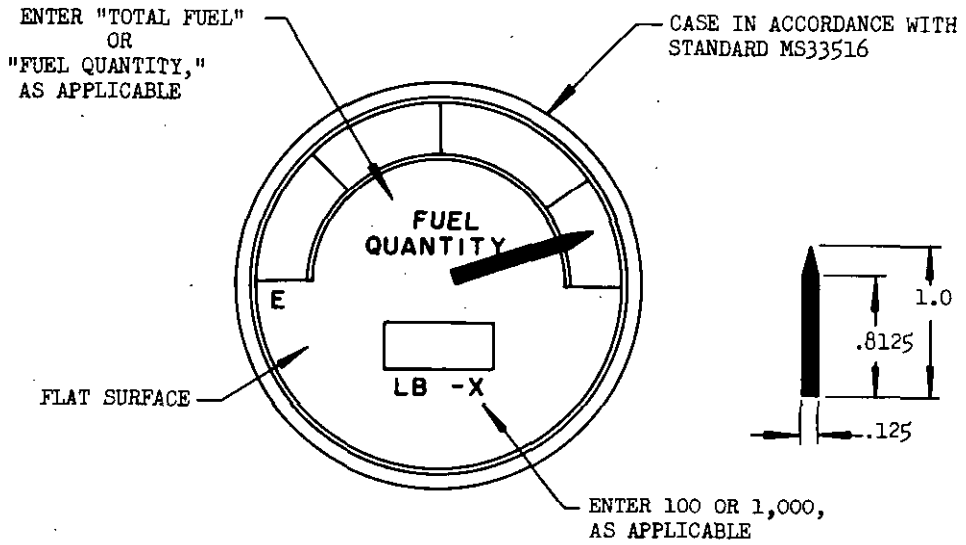
POINTER

MARKINGS	*HEIGHT OR LENGTH* ±.010	*WIDTH OF LINE OR GRADUATION* ±.005
TOTAL FUEL	.156	.031
LB - X100	.130	.025
MAJOR GRAD	.250	.031
MINOR GRAD	.188	.015
NUMERAL	.200	.031

DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: ANGLES ±0°15', DECIMALS ±0.005.

FIGURE 3. Dial - 2-inch size, 320-degree scale

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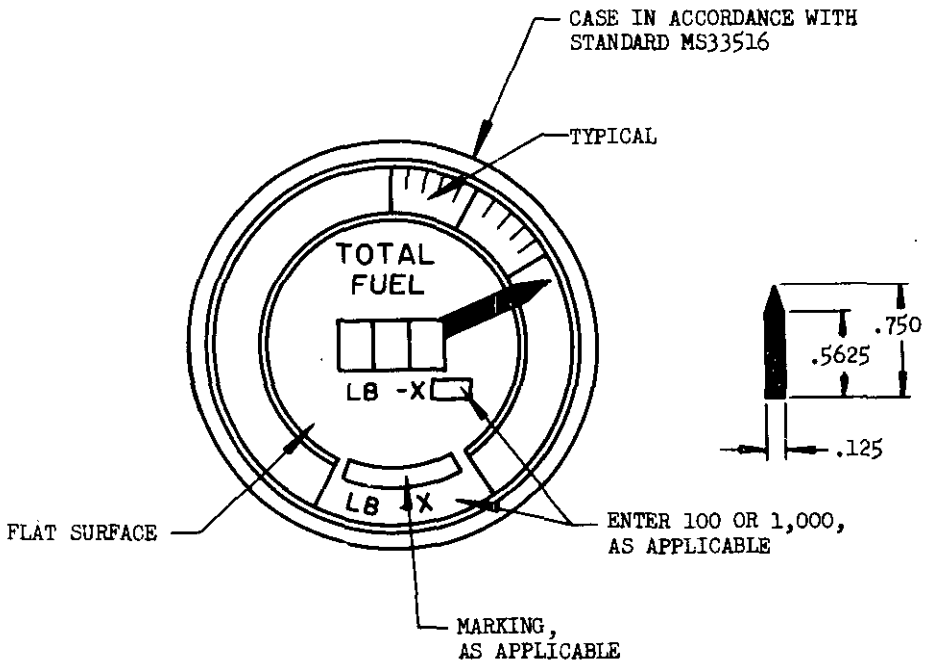


MARKINGS	HEIGHT OR LENGTH ±.010	WIDTH OF LINE OR GRADUATION ±.005
FUEL QUANTITY	.156	.031
TOTAL FUEL	.156	.031
LB - X	.156	.025
E	.200	.031
COUNTER NUMERALS	.250	.031

DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: ANGLES ±0°15', DECIMALS ±0.005.

FIGURE 4. Dial - 3-1/4 inch size, 180-degree scale

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MARKINGS	HEIGHT OR LENGTH ±.010	WIDTH OF LINE OR GRADUATION ±.005
TOTAL FUEL	.156	.031
LB -X	.130	.025
MAJOR GRAD	.3125	.031
MINOR GRAD	.1875	.015
NUMERALS	.250	.031

DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: ANGLES ±0°15', DECIMALS ±0.005.

FIGURE 5. Dial - 3-1/4 inch size, 320-degree scale

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3.4.8.5.3.2.1 Lamp location.- The lamps shall be permanently mounted inside the instrument case. No provision shall be made for field replacement of the lamps. The lamps shall be so installed that replacement may be accomplished at a repair depot. At least three lamps shall be used to illuminate dials which conform to figures 1, 3, 4, and 5, and at least two lamps shall be used for dials conforming to figure 2.

3.4.8.5.4 Cover glass.- The cover glass shall be suitable for hermetic sealing and shall be clear, flat, and free from flaws which interfere with the normal reading of the instrument. The thickness of the cover glass shall be the minimum practicable and entirely suitable for the purpose intended.

3.4.8.5.5 Dial.- The indicator dial shall be securely fastened to the case or to the frame of the mechanism of the indicator by not less than two screws in such manner that it will not turn or loosen when the instrument is vibrated. The dial material shall be subject to the approval of the cognizant military activity. Numerals on the counter drums shall be approximately on the same plane as the numerals on the front dial. Dials conforming to figures 3 and 5 shall present two indications on the same dial. The counter shall indicate total fuel, and the pointer shall indicate fuel in an individual tank(s).

3.4.8.5.5.1 Pointer scale.- The pointer scale of figures 3 and 5 shall be graduated and numbered to indicate quantities in pounds at even intervals. The intervals employed shall be determined by the capacity of the tank and the scale length, in order to obtain a scale having sufficient graduations and numerals for easy reading without overcrowding. The indicating scale shall provide a minimum scale angle of 300 degrees. The pointer scale for figures 1, 2, and 4 shall be 180 degrees.

3.4.8.5.5.2 The pointer scale of figures 3 and 5 shall be marked according to the following procedure:

(a) Indicator dials shall be calibrated in pounds.

(b) The graduations shall be uniformly spaced.

(c) The end point of the calibration shall be number of pounds obtained by multiplying the tank volume in gallons by the end point density specified in 3.4.3. By tank volume is meant that part of the total volume which is measured by the gage.

(d) The scale end shall be the last division before the end point (for example, if the weight of fuel at the end point is 4,480 lb and minor graduations represent 200 lb, the scale end will be the minor graduation representing 4,400 lb).

3.4.8.5.5.3 Dial markings.- All markings shall be durable to withstand usage encountered in service. The dials shall be marked as shown on figures 1, 2, 3, 4, and 5. The pointer scale of figures 3 and 5 shall be linear within ±0° 30'. Wherever practicable, each major graduation shall be identified by a single numeral.

3.4.8.5.6 Pointers.- Pointers shall be light and sufficiently rigid to prevent oscillation under vibration. They shall be firmly attached to the associated mechanism. Pointers shall have the general appearance as shown in figures 1, 2, 3, 4, and 5 with the necessary cross section as required for integral lighting. The pointer shall extend into the scale to a distance of 1/3 to 2/3 of the shortest graduation.

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3.4.8.6 Intermediate device.- The intermediate device (3-unit system) shall include all parts of the gage exclusive of the tank units, the separately mounted compensator sensing unit (if applicable), the indicator, and the external wiring, and may house component parts of the compensator device. The intermediate device shall be designed to provide readily accessible electrical connections for testing the gage system and making calibration adjustments, using substitute precision capacitors. The intermediate device shall incorporate adjustment features to permit universal application without substitution of component parts of the device. No shock mounts shall be used on the intermediate device.

3.4.8.7 Electrical cable.- The gage system shall be suitable for use with external unshielded connectors, cable wire, type RM-22 of Specification MIL-W-7139, and coaxial cable type RG-179/u of Specification MIL-C-8721. Any other wire shall be of equal, or greater, tensile strength and shall have the same diameter as the above wires. Internal hookup wire shall be type B of Specification MIL-W-16878. Electrical cable used within the fuel tank subject to the action of the fuel shall be entirely suitable for the purpose intended. In applications where the temperature may exceed 90°C in compartments of the aircraft structure through which the electrical interconnecting cables will be routed, the cables shall be a type designed for high-temperature use and shall be subject to approval of the cognizant military activity. Internal wiring shall be neat and accomplished in a manner that individual wires may be easily traced. Wires shall be tied to the terminals prior to soldering.

3.4.8.7.1 Cable length.- Accuracy and sensitivity of the gage shall not be adversely affected by the use of the tank unit interconnecting cables within the limits of from 5 to 200 feet.

3.4.8.8 Electrical connectors.- The electrical connectors used, with the exception of those required for the tank unit circuits, shall be in accordance with Specification MIL-C-5015, unless otherwise approved by the cognizant military activity. Coaxial cable connectors shall be in accordance with Specification MIL-C-25516.

3.4.8.8.1 The polarity for the coaxial connectors shall be as follows:

Polarity of Specification MIL-C-25516	Applications
F	High impedance
A	Low impedance
B	Compensators

3.4.8.9 Input electrical power.- The indicator shall be designed to operate from 115V, 400-cps, single-phase aircraft power supply described in Specification MIL-E-7894.

3.4.8.10 Pointer overshoot.- The gage shall be designed to provide steady indication of the pointer. In approaching null, the pointer shall not overshoot the final position more than one time and shall return to the final position within 2 seconds.

3.4.8.11 Calibration adjustment.- Adjustments of the indicator pointer shall be such that a change in the full adjustment does not appreciably affect the position of the indicator pointer at the zero point. The empty and full adjustments shall provide a sensitivity such that adjustments can be easily accomplished under service operations. The design shall be such that the complete gage system including the compensator, can be accurately and easily checked when the gage is installed in the airplane by means of the MD-1 field tester conforming to Specification MIL-T-8579 or MD-2 field tester conforming to Specification MIL-T-4687, as applicable.

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3.4.8.11.1 Two-unit system.- Means shall be provided for adjusting the electrical circuit of the 2-unit system for the calibration of the indicator. The internal empty and full adjustments shall be located on the back of the indicator and not on the tank units and shall be so designed that the complete range of each adjustment shall not cause pointer movement of more than 5 percent of full scale and can be easily accomplished under service operations.

3.4.8.11.2 Three-unit system.- Means shall be provided for adjusting the electrical circuit for range and calibration of the system. The external adjustment shall be located on the intermediate device and not on the tank units.

3.4.8.12 Gage failure indication.- Each indicator shall be provided with a readily accessible terminal connection for use with a remote test switch to provide a means of checking gage operation. Operation of the external test switch shall cause the indicator pointer to rotate counterclockwise from a balance point on any portion of the dial.

3.5 Interchangeability.- All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of Specification MIL-D-5028.

3.6 Screw threads.- Unless otherwise specified, the threads of all machine screws, 0.060 inch or larger in diameter, shall conform to Specification MIL-S-7742.

3.7 Lubrication.- Lubrication shall be accomplished in accordance with Specification MIL-L-6880. Lubricants shall conform to applicable Government specifications, unless otherwise approved by the activity responsible for qualification.

3.8 Finishes and protective coatings.-

3.8.1 Finishes and protective coatings which will crack, chip, or scale during normal service life or are caused by extremes of atmospheric conditions, shall not be used.

3.8.2 Aluminum-alloy parts.- Aluminum-alloy parts, where practicable, shall be covered with an anodic film conforming to Specification MIL-A-8625.

3.8.3 Steel parts.- Steel parts shall be cadmium or zinc plated, where practicable, in accordance with Specification QQ-P-416, type II or III, as applicable, and of a class that is adequate to achieve the degree of protection required, of Specification QQ-Z-325.

3.9 Performance.- The gage shall satisfactorily meet the requirements specified in section 4, when subjected to the applicable tests.

3.10 Identification of product.-

3.10.1 Nameplate.- Nameplates conforming to Specification MIL-P-6906 shall be securely attached to each separately mounted unit of the gage and shall be legibly and permanently marked. The information marked in the spaces provided on the nameplate shall be in accordance with Standard MIL-STD-130.

3.10.2 Use of AN or MIL designations.- AN or MIL designations shall not be applied to a product, except for Qualification test samples, nor referred to in correspondence or sales matter until notification has been received from the activity responsible for qualification.

3.11 Workmanship.-

3.11.1 General.- The gage, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, welding and brazing, painting, riveting, machine-screw assemblies, and freedom of parts from burrs and sharp edges.

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3.11.2 Screw assemblies.- Assembly screws and bolts shall be tight. The word "tight" means that the screw or bolt cannot be appreciably tightened further, without damage or injury to the screw or bolt or threads.

3.11.3 Riveting.- Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.11.4 Gears.- Gear assemblies shall be properly alined and meshed and shall be operable without interference, tight spots, loose spots, or other irregularities. Where required, gear assemblies shall be free from backlash.

3.11.5 Cleaning.- The gage shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign material, after final assembly. Burrs and sharp edges, as well as resin flash that may crumble, shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of tests.- The inspection and testing of gages shall be classified as follows:

(a) Qualification tests: Qualification tests are those tests performed on samples submitted for approval as qualified products.

(b) Acceptance tests: Acceptance tests are those tests performed on individual lots which have been submitted for acceptance.

4.2 Qualification tests.-

4.2.1 Sampling instructions.- The Qualification test samples shall consist of two complete gage systems with each containing the following components:

(a) One tank unit having a capacitance distribution as specified in table III, tank unit No. 1.

(b) One tank unit having a capacitance distribution as specified in table III, supplemental tank unit.

(c) One indicator of each type upon which qualification is sought as described in 4.2.1.1.

(d) One totalizer as described in 4.2.1.2.

4.2.1.1 Indicator.- The pointer scale of figures 3 and 5 shall be graduated from 0 to 100-percent. The 100 percent graduation shall be at approximately 300 degrees. The percent reading for complete immersion in nominal full shall be as defined herein and specifically in 6.4.5. Indicators conforming to figures 3 and 5 submitted for qualification shall have a maximum end point total fuel capacity of 20,000 pounds. Indicators conforming to figure 2 shall be type A; the full scale indication of the pointer shall correspond to a counter indication of 8,000 pounds. On indicators conforming to figures 1 and 4, a full scale indication on the pointer shall correspond to a counter indication of 500,000 pounds.

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TABLE III

Characterization of tank units for Qualification tests

Tank unit No. 1					Supplemental tank unit		
Distance of fuel level below top of sensing portion (inches)	Dry capacitance per unit length (relative units)	Increase over dry capacitance (percent of total)	Indicator reading (percent full scale JP-4 only) 1/	Indicator reading (percent full scale JP-4 and octane) 2/	Distance of fuel level below top of sensing portion (inches)	Dry capacitance per unit length (relative units)	Increase over dry capacitance (percent of total)
0	3.5	100.0	90.93	87.57	0	0.4	100.0
1	3.5	96.5	87.75	84.51	2	0.8	98.8
2	3.5	93.0	84.56	81.44	4	1.2	96.8
3	3.5	89.5	81.38	78.38	6	1.6	94.0
4	3.5	86.0	78.20	75.31	8	2.0	90.4
5	3.5	82.5	75.02	72.25	10	2.4	86.0
6	3.5	79.0	71.83	69.18	12	2.8	80.8
7	3.5	75.5	68.65	66.12	14	3.2	74.8
8	3.5 - 7.5	72.0	65.47	63.05	16	3.6	68.0
9	77.125	64.69	58.82	56.65	18	4.0	60.4
10	6.750	57.75	52.51	50.57	20	3.6	52.8
11	6.375	51.19	46.55	44.83	22	3.2	46.0
12	6.000	45.00	40.92	39.41	24	2.8	40.0
13	5.625	39.19	35.64	34.32	26	2.4	34.8
14	5.250	33.75	30.69	29.55	28	2.0	30.4
15	4.875	28.69	26.09	25.12	30	1.6	26.8
16	4.500	24.00	21.82	21.02	32	1.2	24.0
17	4.125	19.69	17.90	17.24	34	0.8	22.0
18	3.750	15.75	14.32	13.79	36	0.4	20.8
19	3.375	12.19	11.08	10.67	38	0.4	20.0
20	3.000	9.00	8.18	7.88	40	0.4 - 2.4	19.2
21	2.625	6.19	5.63	5.42	42	2.4	14.4
22	2.250	3.75	3.41	3.28	44	2.4	9.6
23	1.875	1.69	1.54	1.48	46	2.4	4.8
24	1.500	0	0	0	48	2.4	0

1/ This column lists the required indicator readings when the tank unit is connected into the gage circuit and the sensing portion of the tank unit is immersed to the specified depths in nominal fuel of dielectric constant 2.090.

2/ This column lists the required indicator readings when the tank unit is connected into the gage circuit and the sensing portion of the tank unit is immersed to the specified depths in nominal fuel of dielectric constant 2.041.

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4.2.1.2 Totalizer.- Totalizers submitted for qualification approval shall be designed to combine the indications of three gages corresponding to relative tank volumes of 6:3:1. The requirements for the totalizer indicator shall be the same as those for the gage indicator. Totalizers shall not be submitted for qualification unless approval has been previously authorized for individual gages associated with the totalizer, or the individual gages are submitted jointly with the totalizers for qualification.

4.2.1.2.1 Gage outputs for totalizer.- When submitted in conjunction with totalizers, the gages shall be provided with means for supplying signals suitable for application to the totalizers.

4.2.1.2.2 Totalizer inputs.- The manufacturer shall furnish with the totalizers a set of three gages, provided with suitable calibration charts to serve as inputs for testing the totalizers. In lieu of the three gages, the manufacturer may submit artificial devices simulating the gages. If artificial devices are submitted, the manufacturer shall provide sufficient data to verify that such devices are equivalent to the respective individual gages.

4.2.2 Qualification required.- Prior to actual procurement, the product which this specification covers shall be as follows:

(a) Pass the Qualification tests specified herein, conducted by or under the surveillance of a cognizant military activity.

(b) If the product is later modified in any way, the modified form shall be subjected to and shall pass the same Qualification tests.

4.2.2.1 Wiring diagrams.- The manufacturer shall submit to the activity responsible for qualification complete wiring diagrams of all the equipment on which qualification is desired. The wiring diagrams shall include values of components.

4.2.2.1.1 Bill of materials.- The manufacturer shall submit the type, numerical ratings (if applicable), and manufacturer's name of the following components: capacitors, relays, switches, amplifying devices, rectifying devices, and temperature-compensation devices. For example: "Capacitor, ceramic, 10 uuf, MANCO Corporation."

4.2.2.2 Calibration data for Qualification tests.- The gage manufacturer shall furnish the activity responsible for qualification, on 8 by 10-1/2 inch paper, two copies of calibration data containing the following information:

(a) Data for both the gage tank unit and the supplemental tank unit.

(1) Distance from bottom surface of mounting flange to top of sensing portion and distance from bottom of sensing portion to bottom of tank unit. (Length of nonsensing portion at both ends of tank unit for internally mounted units.)

(2) Dry capacitance.

(3) Total capacitance for complete immersion in nominal fuel.

(b) Data for compensator sensing unit.

(1) Dry capacitance.

(2) Capacitance for immersion in nominal fuel to levels specified in first column of table III. (If compensator sensing unit is a separate unit, its location relative to the gage tank unit shall be specified.)

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4.2.3 Tests.- The Qualification tests shall consist of all the tests of this specification, as described under "Test methods" and shall be tested in the sequence as listed. Samples shall be identified with the manufacturer's own part number and as required, and forwarded to the activity responsible for qualification, designated in the letter of authorization from that activity. (See 6.3.)

4.3 Acceptance tests.- The Acceptance tests shall consist of Individual tests and Sampling tests.

4.3.1 Individual tests.- The Individual tests shall consist of the following tests:

(a)	Examination of product	4.5.1
(b)	Indicator operation at room temperature	4.5.2
(c)	Tank unit dry capacitance test	4.5.3.1
(d)	Compensator sensing unit dry capacitance test	4.5.3.2
(e)	Tank unit and compensator sensing unit leakage	4.5.4
(f)	Tank unit calibration	4.5.5
(g)	Sealing test	4.5.6
(h)	Dielectric strength	4.5.7

4.3.2 Sampling tests.-

4.3.2.1 Sampling plan A tests.- Five indicators which have been manufactured under essentially the same conditions shall be selected at random from each 500 or fraction thereof produced on the order and subjected to the following tests as described under "Test methods." For tank units and compensator sensing units, one unit shall be selected at random from each 100 or fraction thereof produced on the order and subjected to the following tests as described under "Test methods":

(a)	Individual tests	4.3.1
(b)	Complete gage system scale error at room temperature	4.5.8
(c)	Compensation test	4.5.9
(d)	Indicator scale error at room temperature	4.5.10
(e)	Tank unit assembly and compensator sensing unit dry capacitance	4.5.11
(f)	Voltage and frequency variation	4.5.12
(g)	Indicator scale error at low temperature	4.5.13
(h)	Tank unit and compensator sensing unit capacitance test at low temperature	4.5.14
(i)	Indicator scale error at high temperature	4.5.15
(j)	Tank unit and compensator sensing unit capacitance test at high temperature	4.5.16
(k)	Vibration error	4.5.17

4.3.2.2 Sampling plan B tests.- Unless otherwise specified, two complete gages selected at random from the first 10 items of the contract or order shall be subjected to the following tests, as described under "Test methods":

(a)	Sampling plan A tests	4.3.2
(b)	Speed of response	4.5.18
(c)	Radio noise suppression	4.5.19
(d)	Indicator magnetic effect	4.5.20
(e)	Extreme low-temperature exposure	4.5.21
(f)	High-temperature exposure	4.5.22
(g)	Tank unit water immersion test	4.5.23
(h)	Humidity	4.5.24
(i)	Tank unit fuel resistance tests	4.5.29
(j)	Fungus resistance	4.5.25
(k)	Salt spray	4.5.26
(l)	Indicator cycling test	4.5.27
(m)	Thermal shock	4.5.28

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4.3.2.3 Rejection and retest.- When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined.

4.3.2.3.1 Individual tests may continue.- For operational reasons, Individual tests may be continued pending the investigation of a Sampling test failure. But final acceptance of items on hand or later produced shall not be made until it is determined that items meet all the requirements of the specification.

4.3.3 Defects in items already accepted.- The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the cognizant military activity of all defects likely to be found and methods of correcting them.

4.3.4 Miscellaneous equipment tests.- At least 3 months prior to fabrication of the first new airplane type the following components of the fuel quantity system shall be submitted to the cognizant military activity for testing against the general requirements of this specification:

(a) All capacitance simulators.

(b) Ratio devices (if separate).

(c) Relays.

(d) Switches.

(e) A series of tank units representing (1) each type of nonqualified mounting used in the aircraft (2) the longest and shortest nonqualified tank unit, (3) both compensated and uncompensated types, (4) separate nonqualified-compensator sensing units if used, and (5) all tank units employing level sensing devices, such as thermistors with the sensing devices installed.

The cognizant military activity reserves the right to waive these tests.

4.4 Test conditions.-

4.4.1 Standard atmospheric conditions.- Standard conditions shall be as specified in the paragraph titled "Standard Conditions" of Specification MIL-E-5272. When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.4.2 Position.- Unless otherwise specified, the indicator shall be tested with the dial in the normal operating position.

4.4.3 Test power.- Unless otherwise specified herein, the gages shall be tested with 115V 400-cps single-phase a-c power supply. The voltage and frequency tolerance shall be ±2 percent.

4.4.4 Master test standards.-

4.4.4.1 Test instrument.- Whenever a master test instrument is specified for tests of gages, it shall be in the form of a precision variable capacitor or precision fixed capacitors. The accuracy of the precision capacitors, fixed or variable, at 25°C, shall be 0.2 percent for capacitances in excess of 50 uuf, and 0.1 uuf for capacitances less than 50 uuf. For tests of totalizers, the master test instrument shall consist of a suitable number of gages, or equivalent input devices, which have been calibrated to an accuracy of 0.2 percent.

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4.4.4.2 Capacitance bridge.- The master capacitance bridge specified herein shall be used to measure the capacitance of tank units, compensator sensing units, and tank unit simulators. The accuracy of the bridge at an ambient temperature of 25°C shall be 0.2 percent for capacitances in excess of 50 uuf, and 0.1 uuf for capacitances less than 50 uuf.

4.4.5 In all subsequent tests and measurements specified herein, the tolerance of the measuring equipment shall be added to the specified tolerance for the given test or measurement.

4.5 Test methods.-

4.5.1 Examination of product.- Each gage shall be examined to determine compliance with the requirements specified herein with respect to materials, workmanship, and marking.

4.5.2 Indicator operation at room temperature.- All individual and totalizer indicators identified by the same manufacturer's part number shall be individually checked against fixed design calibration data to determine:

(a) That the scale error at each major graduation is within ±1/2 percent of full scale. The scale error of each indicator shall be determined under conditions of both increasing and decreasing readings.

(b) That the counter and pointer will travel from zero to full scale indication in less than 90 seconds and more than 15 seconds.

(c) That, where applicable, the output signal at each major graduation is within ±1/2 percent of the fixed design value at full scale. This test is required only for those indicators providing signals for totalizers.

(d) That level switching mechanism and other items, such as potentiometers, incorporated in the design for specific applications operate properly.

(e) That the pointer moves smoothly over the complete range of the indicator.

(f) That all external adjustments function correctly.

(g) That the dead spot does not exceed 0.2 percent at any two points selected by the Inspector. The dead spot shall be determined as follows: The input capacitance of the gage system shall be slowly varied by means of the master test instrument, first in one direction and then in the other. The capacitance change required to produce indicator rotation shall be noted and shall be computed as a percentage of the total input capacitance range for which the gage is calibrated; this percentage is defined as the dead spot. For totalizers, a suitable input signal provided by a master test instrument shall be employed in lieu of input capacitance.

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4.5.3 Tank unit and compensator sensing unit electrical capacitance.-

4.5.3.1 Tank unit dry capacitance.- The dry capacitance of each tank unit shall be determined by means of the master capacitance bridge. The capacitance value established shall not differ from the respective value specified by the manufacturer by more than 1.0 percent, or 0.5 uuf, whichever is greater. This test may be combined with the Tank unit calibration test specified in 4.5.5.

4.5.3.2 Compensator sensing unit dry capacitance.- The dry capacitance of each compensator sensing unit shall be determined by means of the master capacitance bridge. The capacitance value established shall not differ from the value specified by the manufacturer by more than 1.5 percent, or 0.5 uuf, whichever is greater.

4.5.4 Tank unit and compensator sensing unit leakage.- Tank units and compensator units of the externally mounted type shall be installed in a suitable pressure chamber equipped with the proper size flange fittings. A pressure of 30 ±5 psi shall be applied within the pressure chamber, and no leak shall be detected on the outer side of the tank unit mounting flange and connector head assembly. A suitable liquid shall be used for detecting the presence of air bubbles. The duration of this test shall be 5 minutes. A leak shall be defined as a periodic bubbling from a specific area at a frequency of at least one bubble each minute.

4.5.5 Tank unit calibration.- The dry capacitance of each of the tank units shall be determined by means of the master capacitance bridge. Each of the tank units shall be individually immersed in a representative fuel sample of the levels specified in table III or IV, as applicable, and the added capacitance, attributable to the fuel, shall be determined. The added capacitance values, expressed as percentages of the total added capacitance for complete immersion, shall agree within 1.5 percent with the values specified in table III or IV, as applicable.

TABLE IV

Individual tank unit calibration data 1/

Height of fuel level in inches from bottom of tank unit	Added capacitance attributable to fuel, expressed in percent of total added capacitance for complete immersion		
	Tank unit No. 1	Tank unit No. 2	Additional tank units, as applicable

1/ See 6.2.1(j).

4.5.5.1 The test specified in 4.5.5 shall be conducted on the 1st and every 10th tank unit thereafter on the contract or order. In the event a tank unit fails to meet the specified requirements, it shall be rejected. When a tank unit failure occurs, the next 10 consecutive tank units manufactured shall be subjected to the test specified in 4.5.5.

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4.5.6 Sealing test.- Hermetically sealed components shall be tested for leaks by means of a mass spectrometer type of helium leak detector. Where a nitrogen-helium mixture is utilized as the filling medium, the detected leak rate shall not exceed 0.01-micron cubic foot per hour at a pressure differential of 1 atmosphere. Where 100 percent helium is utilized as the filling medium, the leak rate shall not exceed 0.1-micron cubic foot per hour.

4.5.7 Dielectric strength.- There shall be no breakdown of insulation of any of the equipment when it is subjected for 1 minute to a test voltage of 500V rms ac at a commercial frequency applied between any pair of points desired by the testing activity. If the design of the gage is such that there exists between the test points a number of electrical circuit elements which form a conducting path for alternating current which would be damaged by the test voltage, these elements shall be removed prior to application of the test voltage. Where hermetically sealed enclosures are employed, the above test shall be conducted prior to sealing the enclosures. After the enclosures are sealed and pressurized, the same test shall be repeated with the exception that the voltage shall be reduced to 200V and shall not be conducted between points where circuit elements would be damaged by the test voltage.

4.5.8 Complete gage system scale error at room temperature.- The complete gage system shall be electrically connected, and the tank unit(s) shall be installed in a suitable test chamber. The external compensator sensing unit, if used, shall be installed in the correct position with the tank unit(s) in the test chamber. The gage shall be set to read "zero" with the dry tank unit(s) and the compensator sensing unit, and set to the added capacitance values specified for complete immersion in nominal fuel, and the gage shall be set to read "nominal full." The master test instruments shall be removed, and nominal fuel shall be poured into the test chamber to the levels specified in table III or V, as applicable, and the indicator readings shall be noted. The difference between the observed readings and the readings specified shall not exceed 2.0 percent of indication +1.0 percent of full scale indication.

TABLE V
Gage system calibration data 1/

Height of fuel level in inches from bottom of lowest tank unit	Total tank unit capacitance for nominal fuel, uuf	Capacitance of compensator sensing unit for nominal fuel, uuf	Indicator reading in pounds	Output voltage or resistance, or both, applicable to gages which are a part of a totalizing gage system

1/ See 6.2.1(j).

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4.5.8.1 In lieu of nominal fuel, any suitable fuel may be used whose dielectric constant has a known value between 2.00 and 2.10. If the dielectric constant differs from the nominal value, the specified indicator readings shall be multiplied by the correction factor listed in table VI, VII, or VIII, corresponding to the actual dielectric constant of the fuel.

TABLE VI

Compensator calibration table - JP-4 fuel only

Dielectric constant	Indicator reading (percent full scale)	Correction factor	Dielectric constant	Indicator reading (percent full scale)	Correction factor
1.85	75.34	0.8286	2.06	89.06	0.9794
1.86	76.03	0.8362	2.07	89.67	0.9862
1.87	76.71	0.8437	2.08	90.28	0.9929
1.88	77.40	0.8512	2.09	90.93	1.0000
1.89	78.07	0.8586	2.10	91.50	1.0063
1.90	78.74	0.8660	2.11	92.11	1.0130
1.91	79.41	0.8734	2.12	92.71	1.0196
1.92	80.07	0.8807	2.13	93.31	1.0262
1.93	80.74	0.8880	2.14	93.91	1.0328
1.94	81.40	0.8953	2.15	94.50	1.0393
1.95	82.06	0.9024	2.16	95.09	1.0458
1.96	82.71	0.9097	2.17	95.67	1.0522
1.97	83.36	0.9168	2.18	96.26	1.0587
1.98	84.00	0.9239	2.19	96.85	1.0651
1.99	84.65	0.9310	2.20	97.42	1.0714
2.00	85.29	0.9380	2.21	98.00	1.0778
2.01	85.93	0.9450	2.22	98.57	1.0841
2.02	86.55	0.9519	2.23	99.14	1.0903
2.03	87.19	0.9589	2.24	99.72	1.0967
2.04	87.81	0.9657	2.245	100.00	1.0998
2.05	88.43	0.9726	2.25	100.28	1.1029

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TABLE VII

Compensator calibration table - JP-4 and octane fuels

Dielectric constant	Indicator reading (percent full scale)	Correction factor	Dielectric constant	Indicator reading (percent full scale)	Correction factor
1.85	74.70	0.8527	2.05	88.17	1.0065
1.86	75.40	0.8607	2.06	88.81	1.0138
1.87	76.10	0.8687	2.07	89.45	1.0211
1.88	76.80	0.8767	2.08	90.09	1.0284
1.89	77.49	0.8846	2.09	90.72	1.0356
1.90	78.18	0.8925	2.10	91.35	1.0428
1.91	78.87	0.9003	2.11	91.98	1.0500
1.92	79.55	0.9081	2.12	92.61	1.0572
1.93	80.23	0.9159	2.13	93.23	1.0643
1.94	80.92	0.9237	2.14	93.85	1.0714
1.95	81.59	0.9314	2.15	94.48	1.0785
1.96	82.26	0.9390	2.16	95.09	1.0855
1.97	82.93	0.9467	2.17	95.70	1.0925
1.98	83.60	0.9543	2.18	96.31	1.0994
1.99	84.25	0.9618	2.19	96.92	1.1064
2.00	85.15	0.9720	2.20	97.53	1.1133
2.01	85.58	0.9769	2.21	98.12	1.1201
2.02	86.23	0.9843	2.22	98.73	1.1220
2.03	86.88	0.9918	2.23	99.32	1.1338
2.04	87.53	0.9992	2.24	99.92	1.1406
2.045	87.60	1.0000	2.25	100.50	1.1473

TABLE VIII

Correction factors for complete gage scale error at room temperature for the uncompensated system, using octane fuels only

Dielectric constant	Correction factor
1.90	0.9091
1.91	0.9192
1.92	0.9293
1.93	0.9394
1.94	0.9495
1.95	0.9596
1.96	0.9697
1.97	0.9798
1.98	0.9899
1.99	1.0000
2.00	1.0101
2.01	1.0202
2.02	1.0303
2.03	1.0404
2.04	1.0505
2.05	1.0606
2.06	1.0707
2.07	1.0808
2.08	1.0909

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4.5.9 Compensation.- With the empty and full adjustments set as in the Complete gage system scale error at room temperature test specified in 4.5.8, the tank unit(s) and compensator sensing unit shall be completely immersed in each of two fuels, one having a known dielectric constant less than 1.95, and the other having a known dielectric constant greater than 2.15. The difference between the two readings shall agree with the corresponding difference obtained from table VI or VII within 1 percent of full scale indication.

4.5.10 Indicator scale error at room temperature.- The indicator shall be assembled and connected to master test instruments in lieu of the tank unit(s) and compensator sensing unit. The master test instruments shall be set respectively to the specified capacitance values corresponding to empty tank and complete immersion in nominal fuel, and the indicator shall be adjusted to read "zero" and "nominal full." Once set, the adjustments shall not be altered during the remainder of the tests. The master test instruments shall then be set to the capacitance values specified in table V. The observed readings shall be recorded. Scale errors shall not exceed 1/2 percent of full scale indication at any of the test points. Where applicable, the output signals for the totalizer shall be recorded for each test point. The values established shall not differ from the specified values by more than 1/2 percent. The dead spot of the gage shall be determined and shall not exceed 0.2 percent of total input capacitance range at any two points selected by the Inspector.

4.5.10.1 The totalizer shall be tested by means of a master test instrument. The observed readings shall be recorded. Scale errors shall not exceed 1/2 percent of full scale indication at any test point, and the dead spot shall not exceed 0.2 percent of total input range.

4.5.10.2 Reference values.- The observed readings in 4.5.10 and 4.5.10.1 will be referred to hereinafter as the "reference values."

4.5.11 Tank unit assembly and compensator sensing unit dry capacitance.-

4.5.11.1 Tank unit assembly dry capacitance.- The dry capacitance of the assembled tank units providing a single indication shall be measured at room temperature with the master capacitance bridge. The value obtained shall agree with the specified value within 1.0 percent.

4.5.11.2 The dry capacitance of the compensator sensing unit determined in the Compensator sensing unit dry capacitance test specified in 4.5.3.2 shall be recorded.

4.5.11.3 Reference capacitance.- The observed readings in 4.5.11.1 and 4.5.11.2 will be referred to hereinafter as "reference capacitance."

4.5.12 Voltage and frequency variation.- The completely assembled system shall be tested at three different positions on the indicator dial, approximately 5, 50, and 90 percent of maximum indication, under the following voltage and frequency combinations:

	A-c voltage	Frequency
(a)	115	400
(b)	105	360
(c)	125	440

The change in indication from that observed under condition (a) shall not exceed 0.5 percent of full scale indication. The dead spot of the gage shall be determined at midscale indication under conditions (b) and shall not exceed 0.2 percent of total input capacitance range.

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4.5.13 Indicator scale error at low temperature.- The indicator shall be placed in a temperature chamber and subjected to a temperature of -55° ±2°C for a minimum period of 4 hours. The master test instrument shall be maintained at room temperature. At the end of this period, and while the unit is still at the low temperature, the indicator shall be electrically energized for 3 minutes. Scale readings shall be determined by means of the master test instrument. The scale readings shall not differ from the "reference values" by more than 0.75 percent of full scale indication. This test and other tests hereinafter specified covering the same equipment shall apply to indicators associated with gages (with and without the totalizing feature) and totalizers.

4.5.13.1 When the unit has returned to room temperature, scale readings shall again be determined by means of the master test unit. The scale readings shall not differ from the "reference values" by more than 0.5 percent of full scale indication.

4.5.14 Tank unit and compensator sensing unit capacitance test at low temperature.- The tank unit assembly, including the compensator sensing unit, shall be placed in a temperature chamber and subjected to a temperature of -55° ±2°C for a minimum period of 4 hours. At the end of this period, and while the tank unit assembly is still at this temperature, the dry capacitance shall be determined and shall not differ from the respective "reference capacitance" by more than 1.0 percent for tank units and compensator sensing units.

4.5.14.1 When the assembly has returned to room temperature, the dry capacitance shall again be determined and shall not differ from the respective "reference capacitance" by more than 0.5 percent.

4.5.15 Indicator scale error at high temperature.- The indicator shall be placed in a temperature chamber and subjected to a temperature of 70° ±2°C for a minimum period of 4 hours, except that the intermediate device of the 3-unit system shall be subject to a temperature of 100° ±3°C for a minimum period of 4 hours. The units shall be electrically energized to read full scale indication throughout the complete 4-hour period. The master test instrument shall be maintained at room temperature. At the end of this period, and while the unit is still at the high temperature, scale readings shall be determined by means of the master test instrument. The scale readings shall not differ from the "reference values" by more than 0.75 percent of full scale indication.

4.5.15.1 After the unit has returned to room temperature, scale readings shall again be determined by means of the master test instrument. The scale readings shall not differ from the "reference values" by more than 0.5 percent of full scale indication.

4.5.16 Tank unit and compensator sensing unit capacitance test at high temperature.- The tank unit assembly, including the compensator sensing unit, shall be placed in a temperature chamber and subjected to a temperature of 125° ±3°C for a minimum period of 4 hours. At the end of this period, and while the tank unit assembly is still at this temperature, the dry capacitance shall be determined and shall not differ from the respective "reference capacitance" by more than 1.0 percent for tank units and compensator sensing units.

4.5.16.1 When the assembly has returned to room temperature, the dry capacitance shall again be determined and shall not differ from the respective "reference capacitance" by more than 0.5 percent.

4.5.17 Vibration error.- The vibration error test shall be conducted on the components of the gage as specified below.

4.5.17.1 Indicator vibration resonance.- The indicator shall be electrically connected to a master test instrument and subjected to the Vibration tests, Procedure IV, of Specification MIL-E-5272. The circle diameter shall be 0.009 to 0.011 inch. The maximum double amplitude of the pointer oscillation shall not exceed 0.5 percent of full scale indication, and the pointer variation from its original position shall be not greater than 0.5 percent of full scale indication. The test shall be conducted at an indicator reading of zero and at a minimum of two additional points, as selected by the Inspector.

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4.5.17.2 Indicator vibration failure.- The indicator, except the intermediate device of the 3-unit system, shall be electrically connected, as specified in the Indicator vibration resonance test (4.5.17.1), and subjected to the Vibration tests, Procedure V, of Specification MIL-E-5272. Following the 3-hour period, the scale readings shall not differ from the "referenced values" by more than 0.5 percent of full scale indication. The indicator shall be inspected thoroughly for damage or defects resulting from the Vibration tests.

4.5.17.3 Intermediate device vibration failure.- The intermediate device of the 3-unit system shall be subjected to the Vibration tests, Procedure I, of Specification MIL-E-5272 for each of the 3 mutually perpendicular axes at room temperature only for a period of 2 hours.

4.5.17.4 Tank unit assembly vibration.- The tank unit assembly, including the compensator sensing unit, shall be subjected to the frequency range of vibratory accelerations or double amplitudes not exceeding those shown below:

Frequency in cpm	Double amplitude
300 to 600	0.050 inch
600 to 4,500	0.036 inch
4,500 to 30,000	10g

4.5.17.4.1 Tank unit resonance.- All parts of the gage shall be electrically connected to each other and to the master test instrument. The gage shall be adjusted to read approximately "midscale" by means of the master test instrument. The tank units and the compensator sensing unit shall be mounted as a separate item with the flange (for flange-mounted units) fixed to the vibration stand, and with end supports and center supports substantially in accordance with types being provided in the respective airplane application. The other types of tank units shall also be mounted on the test stand in a manner similar to that being followed in the respective airplane installation. The resonant frequencies of the tank unit(s) shall be determined by varying the frequency of applied vibration slowly through the range of frequencies at vibratory accelerations not exceeding those specified. This procedure shall be followed successively for vibration applied along the 3 mutually perpendicular axes of the test specimen. The test specimen shall be vibrated at the indicated resonant conditions for 2 hours (time specified refers to one axis of vibration) and with the applied double amplitude or vibratory acceleration specified in 4.5.17.4. These periods of vibration shall be accomplished in sequence for each of the 3 mutually perpendicular axes of vibration. When more than one resonant frequency is encountered with vibration applied along any one axis, the test period shall be accomplished at the most severe resonance frequency. When resonant frequencies are not apparent within the specified frequency range, the specimen shall be vibrated twice as long as those specified for resonance at a frequency of 3,300 cpm and an applied double amplitude of 0.060 inch. While the vibration is being conducted, the maximum double amplitude of the indicator pointer oscillation and the pointer variation from its original setting shall be noted to determine that they do not exceed 0.5 percent of full scale indication. The dry capacitance shall be determined after the test, and shall not differ from respective reference capacitance by more than 0.5 percent. The tank and compensator unit or units, shall be thoroughly inspected to determine that no damage has occurred as a result of the test.

4.5.18 Speed of response.- After the indicator has been maintained at -55°C for a period of at least 2 hours, the time required for the pointer to travel from zero to the end point or vice versa shall not exceed 3 times the time measured for the Indicator operation at room temperature test specified in 4.5.2. The time period shall be measured from the time the system is energized.

4.5.19 Radio noise suppression.- Radio noise suppression shall be in accordance with Specification MIL-I-6181, except that a frequency range of from 0.15 to 150 megacycles shall apply for the radiated radio noise test.

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4.5.20 Indicator magnetic effect.- The indicator, not operating, shall be rotated in a vertical plane about a short bar magnet compass with the nearest part of the indicator 5-1/2 inches from and magnetically east or west of the center of the compass. Starting directly under the compass, the indicator shall be held in positions 0 degrees, 45 degrees, 90 degrees, 135 degrees, 180 degrees, 225 degrees, 270 degrees, and 315 degrees from the initial position. At each of these positions, the indicator shall be rotated on its own horizontal axis until it is in its normal upright position. The horizontal magnetic field intensity shall be 0.17 to 0.19 oersted. The deflection of the compass at any of the specified positions shall not exceed 5 degrees. The test shall be repeated with the indicator operating at rated voltage.

4.5.21 Extreme low-temperature exposure.-

4.5.21.1 Indicator scale error.- The indicator shall be tested as specified in 4.5.13, except that the unit shall be maintained at a temperature of -65° ±2°C for a period of 48 hours, and the temperature raised to -54° ±2°C for a period of 24 hours. During the last 4 hours of the 24-hour period at -54°C, the pressure shall be reduced to 0.82-inch Hg absolute, or less. At the completion of the 24-hour period, and while at the temperature and pressure specified above, the indicator shall be tested to determine that the scale readings shall not differ from the "reference values" by more than 1.0 percent of full scale indication.

4.5.21.1.1 When the indicator has returned to room temperature, scale readings shall again be determined by means of the master test instrument. The scale readings shall not differ from the "reference values" by more than 0.5 percent of full scale indication.

4.5.21.2 Tank unit and compensator sensing unit.- The tank unit assembly, including the compensator sensing unit, shall be tested as described in 4.5.14, except that the assembly shall be maintained at a temperature of -65° ±2°C for a period of 48 hours, and the temperature then raised to -54° ±2°C for a period of 24 hours. During the last 4 hours of the 24-hour period at -54°C, the pressure shall be reduced to 0.82-inch Hg absolute, or less. At the completion of the 24-hour period, and while at the temperature and pressure specified above, the dry capacitance shall be measured and shall not differ from the respective "reference capacitance" by more than 1.0 percent.

4.5.21.2.1 When the assembly has returned to room temperature, the dry capacitance shall again be determined and shall not differ from the respective "reference capacitance" by more than 0.5 percent.

4.5.22 High-temperature exposure.-

4.5.22.1 Indicator scale error.- The indicator shall be tested as described in the Indicator scale error at high temperature test (4.5.15), except that the unit shall be maintained at the high temperature for a period of 24 hours. While still at this temperature, the scale readings shall not differ from the "reference values" by more than 1.0 percent of full scale indication.

4.5.22.1 When the indicator has returned to room temperature, scale readings shall again be determined by means of the master variable capacitor. The scale readings shall not differ from the "reference values" by more than 0.5 percent of full scale indication.

4.5.22.2 Tank unit and compensator sensing unit.- The tank unit assembly and compensator sensing unit shall be tested as described in the Tank unit and compensator sensing unit capacitance test at high temperature (4.5.16), except that the assembly shall be maintained at the high temperature for a period of 24 hours. While still at this temperature, the capacitance shall not differ from the "reference capacitance" by more than 1.0 percent, respectively.

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4.5.22.2.1 When the assembly has returned to room temperature, the dry capacitance shall again be determined and shall not differ from the "reference capacitance" by more than 0.5 percent, respectively.

4.5.23 Tank unit water immersion test.- The tank unit assembly shall be mounted in a suitable test tank. The diameter of this test tank shall be 5-1/2 inches ±1/2 inch. The tank shall then be filled to approximately three-quarters of its capacity with fuel conforming to Specification MIL-F-5624 or MIL-G-5572, as applicable. After the gage reading and the fuel level have been noted, sufficient distilled water shall be poured into the tank to cover the compensator sensing unit, or to fill approximately one-half the remaining space (as applicable), and the tank shall then be sealed. The compensator sensing unit shall remain submerged in distilled water for 1 minute. The tank shall then be inverted for 1 minute and restored to its normal upright position for 5 minutes. The time required to invert the test tank shall be 5 seconds. This procedure shall be repeated four more times. The water shall be drained from the tank and, if necessary, fuel shall be added to restore the original fuel level. The gage reading shall be noted after 5 minutes, and the change of indication from the original reading shall not exceed 2 percent of full scale indication.

4.5.24 Humidity.- All components of the gage, except those completely hermetically sealed, shall be subjected to the Humidity tests, Procedure I, of Specification MIL-E-5272.

4.5.24.1 Indicator scale error.- One hour after exposure to humidity, scale errors shall be determined by means of the master test instrument. The scale readings shall not differ from the "reference values" by more than 2.0 percent of full scale indication.

4.5.24.1.1 Between 24 and 48 hours after exposure to humidity, scale readings shall again be determined by means of the master test instrument. The scale readings shall not differ from the "reference values" by more than 1.0 percent of full scale indication. There shall be no evidence of corrosion or other deterioration which will affect subsequent operation of the gage.

4.5.24.2 Tank unit and compensator sensing unit.- One hour after exposure to humidity, the dry capacitance shall be determined and shall not differ from the "reference capacitance" by more than 2.0 percent.

4.5.24.2.1 Between 24 and 48 hours after exposure to humidity, the dry capacitance shall again be determined and shall not differ from the "reference capacitance" by more than 1.0 percent. There shall be no evidence of corrosion or other deterioration which will affect subsequent operation of the gage.

4.5.25 Fungus resistance.- All components of the gage, except those completely hermetically sealed, shall be subjected for a test period of 14 days, to the Fungus resistance tests, Procedure I, of Specification MIL-E-5272.

4.5.26 Salt spray.- All parts of the gage which are outside of the fuel cell shall be subjected to the Salt spray tests, Procedure I, of Specification MIL-E-5272 for a period of 50 hours. One hour after completion of the test, the components shall be subjected to the following tests.

4.5.26.1 Indicator scale error.- Scale readings shall be determined by means of the master test instrument. The scale readings shall not differ from the "reference values" by more than 2.0 percent of full scale indication. There shall be no evidence of corrosion or other deterioration which will affect subsequent operation of the gage.

4.5.26.2 Tank unit and compensator sensing unit.- The dry capacitance shall be determined and shall not differ from the "reference capacitance" by more than 2.0 percent. There shall be no evidence of corrosion or other deterioration which will affect subsequent operation of the gage.

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4.5.27 Indicator cycling test.- The indicator shall be electrically connected to the master test instrument (variable capacitor simulating the tank unit assembly). During the cycling procedure, the capacitance shall be continuously varied, in order that the indicator pointer moves from "zero" to "end" point and returns to "zero" within 5 minutes ±1 minute. The total cycling period shall be a minimum of 5,000 cycles and shall be conducted in 12-cpm intervals. After each interval, the indicator shall be deenergized electrically a minimum of 10 minutes between cycling intervals. After the cycling test, a scale error test shall be conducted, and the readings established shall not differ from those established prior to the test by more than 0.50 percent of full scale at any test point. The test shall be applicable to all indicators and intermediate devices identified by the manufacturer by different basic part numbers.

4.5.28 Thermal shock.- Hermetically sealed components of the gage shall be alternately subjected to a total of 8 cycles of the immersion in tap water maintained at 85° ±5°C and 5° ±5°C. The length of the time for each bath immersion shall be 30 minutes; not more than 5 seconds shall elapse between bath immersions. No evidence of moisture penetration or damage to the enclosure shall result from this test. Following this test, the test specimen shall be resubjected to the Sealing test specified in 4.5.6.

4.5.29 Tank unit fuel resistance tests.- One tank unit (and compensator sensing unit if applicable) of each type to be qualified shall be soaked for 7 days in type II fuel of Specification MIL-S-3136 at a temperature of 70°C. At the end of this test, the units shall be visually inspected to determine that no deterioration, damage, or defect has occurred caused by immersion in fuel. Following this test, these units shall be subjected to the Tank unit water immersion test specified in 4.5.23 and the Humidity test specified in 4.5.24.

5. PREPARATION FOR DELIVERY

5.1 Application.- The requirements of section 5 apply only to direct purchases by or direct shipments to the Government.

5.2 Packaging and packing.- The indicator shall be packaged and packed for shipment in accordance with group X of Specification MIL-P-6064, in a type II, class I, container.

5.2.1 Items too large for shipment in a container as described in Specification MIL-P-6064 shall be packaged in accordance with Specification MIL-P-5633.

5.2.2 Tank units.- Each tank unit shall be preserved and packaged in accordance with Specification MIL-P-116, Method II.

5.3 Marking of shipments.- Interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129. The identification shall be composed of the following information listed in the order shown:

Stock No. or other identification number as specified
in the purchase document*

INDICATOR (**) FOR GAGE, FUEL-QUANTITY
CAPACITOR-TYPE, NONVACUUM TUBE

*NOTE: The contractor shall enter the Federal Stock No. specified in the purchase document or as furnished by the procuring activity. When the Federal Stock No. is not provided or available from the procuring activity, leave space therefor and enter the Stock No or other identification when provided by the procuring activity.

**Applicable data to be entered by manufacturer.

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5.3.1 Reinspection date.- Reinspection markings shall be added in accordance with ANA Bulletin No. 405.

6. NOTES

6.1 Intended use.- The fuel quantity gages covered by this specification are intended for use in aircraft to indicate the quantity of fuel (in pounds) contained in fuel tanks.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of applicable specification.

(b) Type of gage required (see 1.2).

(c) The level of packaging and packing desired.

6.2.1 Calibration data for Acceptance tests.- The contractor should furnish the Inspector, on 8 by 10-1/2 inch paper, two copies of calibration data containing the following information:

(a) Type of airplane_____ Series_____ .

(b) Tank or tanks for which calibration applies.

(c) Fuel gage system number _____ Cabling diagram _____.

(d) List of all components of the gage system with the manufacturer's part numbers.

(e) The manufacturer's part numbers, and dry capacitances of all individual tank units.

(f) Data for tank unit assembly:

(1) Dry capacitance.
(2) Total capacitance for complete immersion in nominal fuel.
(3) Added capacitance caused by complete immersion in nominal fuel.

(g) Data for compensator sensing unit:

(1) Dry capacitance.
(2) Total capacitance for complete immersion in nominal fuel
(3) Added capacitance caused by complete immersion in nominal fuel.

(h) Relative position of tank units corresponding to locations in the airplane.

(i) Data for indicator dial calibration:

(1) Pounds of fuel equivalent to full scale indication.
(2) Pounds of fuel equivalent to nominal full indication.

(j) Data to complete tables IV and V.

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(k) Data to complete table IX, as follows:

TABLE IX

Totalizing indicator calibration data

Input signal	Indication in pounds of fuel throughout complete scale

6.3 Provisions for qualification.- With respect to products requiring qualification, awards will be made only for such products as have, prior to the bid opening date, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date.

6.3.1 The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Requests for information pertaining to qualification of products covered by this specification should be addressed to the Bureau of Aeronautics, Navy Department, Washington 25, D. C., the activity responsible for qualification, with a copy to the Commander, Wright Air Development Center, Wright-Patterson Air Force Base, Ohio.

6.4 Definitions.-

6.4.1 Amplitude.- The term "amplitude," as used herein, is defined as the distance from the mean position to either extreme. "Double amplitude" is two times the "amplitude."

6.4.2 Zero point.- "Zero point" is defined as the beginning of the dial graduations, which represents an indication of zero fuel quantity (dry tank unit).

6.4.3 Full scale indication.- "Full scale indication" is defined as the indication corresponding to a full tank of fuel having an end point density as defined herein.

6.4.4 Density.- The density values used in this specification are relative values, defined as the ratio of the apparent mass of fuel to the true volume. The apparent mass is the value obtained by weighing the fuel in air at 15.6°C, 50-percent relative humidity and 760 mm of mercury pressure against brass weights of specific gravity 8.4, no corrections being applied for air buoyancy. The true density of aircraft fuels is about 0.009 lb/gal larger than the apparent density.

6.4.5 Nominal full.- "Nominal full" indication is defined as the indication corresponding to the full tank of nominal fuel as defined herein. The percent of full scale readings for nominal full shall be as follows:

(a)	JP-4	90.9
(b)	JP-4 and octane	87.6
(c)	Uncompensated gage	91.6

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6.4.6 Hermetic seal.- A "hermetic seal" is defined as a perfectly closed and airtight seal made between vitric or metallic materials, or both. A hermetic seal is not intended to include seals accomplished by gaskets.

6.4.7 Unless otherwise specified in section 4, "indicator" implies the indicator and intermediate device for the 3-unit system and indicator only for the 2-unit system.

6.4.8 Cognizant military activity.- Cognizant military activity, as used herein refers to either the Bureau of Aeronautics, Navy Department, Washington 25, D. C., or Wright Air Development Center, Wright-Patterson Air Force Base, Ohio.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:
Navy - Bureau of Aeronautics
Air Force

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8279

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